Project Design Phase-II
Data Flow Diagram & User Stories

Date	05-03-2026
Team ID	
Project Name	Malaria Detection using Machine Learning
Maximum Marks	4 Marks

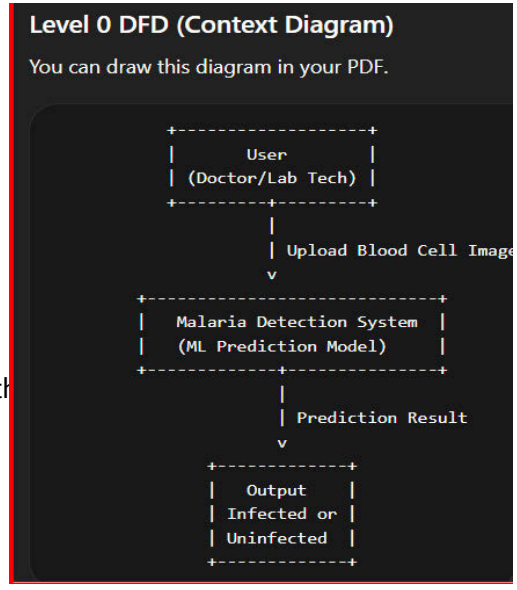
Data Flow Diagrams:

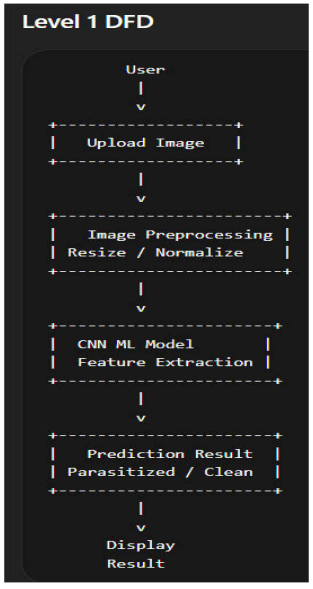
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters the system, how it is processed, where it is stored, and how the results are generated.

In the Malaria Detection System, the user uploads a microscopic blood cell image through the web application. The system processes the image using a trained machine learning model. The model analyzes the features of the blood cell and predicts whether the cell is Parasitized (Malaria infected) or Uninfected (Healthy cell).

The system consists of the following major components:

1. User Interface
2. Allows the user to upload blood cell images.
3. Displays prediction results.
4. 2. Image Processing Module
5. Resizes the image.
6. Normalizes the image for model input.
7. 3. Machine Learning Model
8. A trained Convolutional Neural Network (CNN) analyzes th
9. Extracts important features from the blood cell.
10. 4. Prediction System
11. Classifies the image as Parasitized or Uninfected.
12. 5. Result Display
13. The final output is shown on the web interface.





User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
user	Upload Image	USN-1	As a user, I want to upload a blood cell image.	System accepts image file.	High	Sprint-1
user	Image Processing	USN-2	As a user, I want the system to process the uploaded image.	Image is preprocessed correctly.	High	Sprint-1
user	Malaria Prediction	USN-3	As a user, I want the system to detect malaria from the image.	System predicts infected or uninfected.	Low	Sprint-2
user	View Result	USN-4	As a user, I want to view the prediction result.	Result is displayed clearly.	Medium	Sprint-1
admin	Dataset Management	USN-5	As an admin, I want to manage training data.	Admin can update datasets.	High	Sprint-1
admin	Model Monitoring	USN-6	As an admin, I want to monitor model performance.	Admin dashboard shows accuracy and logs.	Medium	Sprint-2